

Problem

Given a value V , you have an infinite supply of N different coin denominations. Find the minimum number of coins required to make exactly V cents.

You must also print the coins used in an optimal solution.

If it is not possible to make the value V , print -1.

Input Format

The first line contains an integer T , the number of test cases.

For each test case: The first line contains two integers V and N . V — the target value. N — the number of available coin denominations. The second line contains N integers $C[i]$, representing the coin denominations.

Constraints

 $1 \leq T \leq 100$ $1 \leq V \leq 10^6$ $1 \leq N \leq 10^6$ $1 \leq C[i] \leq 10^6$

Output Format

For each test case:

Print the coins used in a minimum-coin solution, separated by spaces. In the next line, print the minimum number of coins. If it is impossible to make V , print -1.

Sample Input 0

```
7 2
2 4
```

Sample Output 0

```
-1
```

Sample Input 1

```
11 4
1 5 6 9
```

Sample Output 1

```
2
```

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Language

Python 3



```
1 V, N = map(int, input().split())
2 coins = list(map(int, input().split()))
3
4 dp = [999999] * (V + 1)
5 dp[0] = 0
6
7 for i in range(1, V + 1):
8     for coin in coins:
9         if coin <= i:
10             dp[i] = min(dp[i], dp[i - coin] + 1)
11
12 if dp[V] == 999999:
13     print(-1)
14 else:
15     print(dp[V])
16
```

Line: 16 Col: 1



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Test against custom input